SOMAIYA VIDYAVIHAR

K J Somaiya Institute of Engineering & Information Technology

K. J. Somaiya Information Technology(KJSIT), Mumbai

Department of Electronics and Telecommunication Engineering

Academic Year 2023-24

PBL I

Presentation

Topic- Obstacle Avoiding Robot

Project Guide –Prof. Amit Kukreja

Project Members: Dhruv Rathod (B45)
Soham Sahastrabuddhe (B47)
Rishabh Tiwari (B58)

Somaiya TRUST

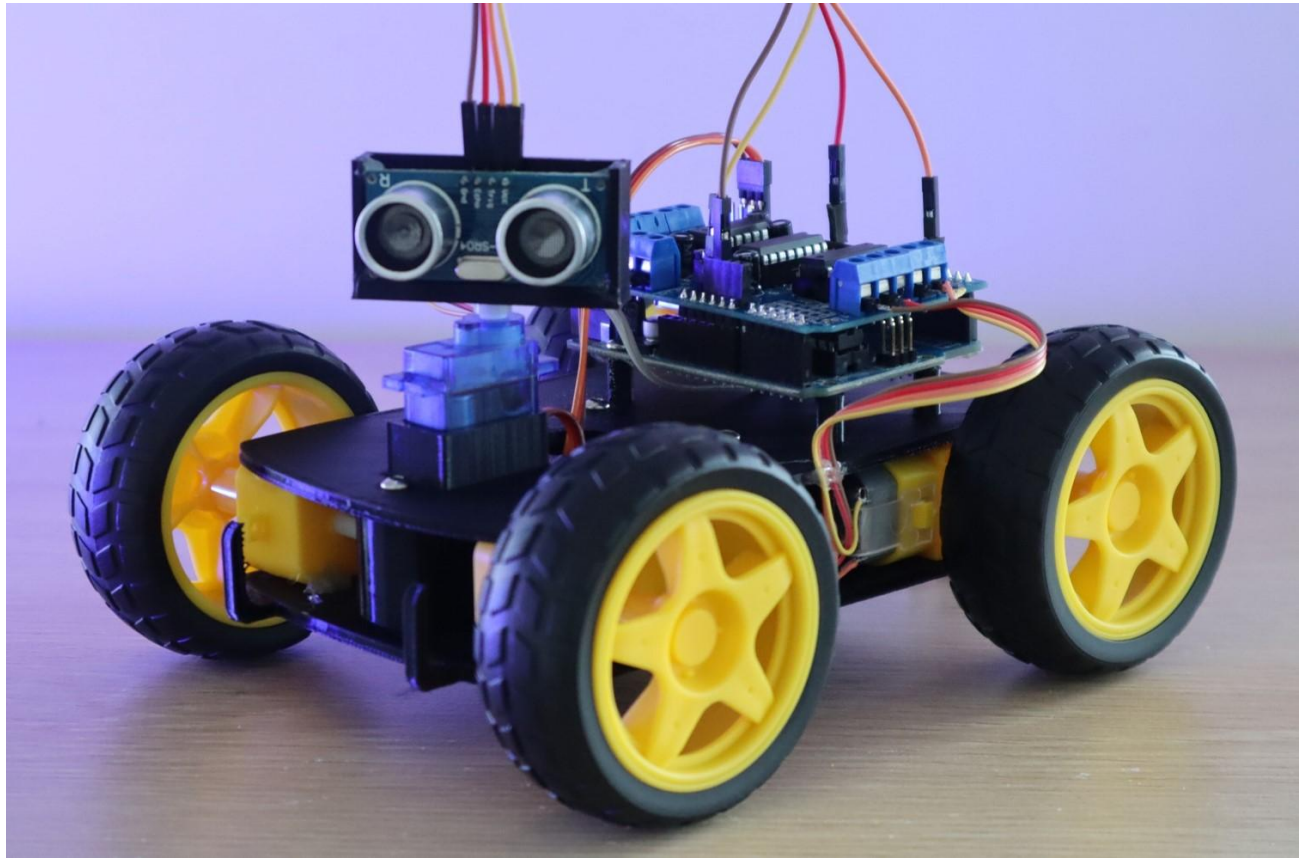
Introduction

- In our journey through automation and robotics, one of the quintessential projects is the obstacle-avoiding bot.

- This project epitomizes the intersection of fundamental robotics and practical application, demonstrating how sensors, microcontrollers, and programming come together to create an autonomous system capable of navigating its environment.

COMPONENTS

- STM32(Black pill)
- Ultrasonic sensor
- DC motor
- Motor driver
- Battery
- Breadboard and Jumper Wires



COST

Sr No.	Components	Price in INR
1	• STM32(Black pill)	1000
2	• Ultrasonic sensor	200
3	• DC motor	150
4	• Motor driver	200
5	• Battery	30
6	• Breadboard and Wires	300

WORKING

- The battery powers the entire bot, including the STM32 microcontroller, motor driver, and ultrasonic sensor.
- It measures the time it takes for the waves to bounce back after hitting an obstacle.
- This time measurement is used to calculate the distance to the obstacle.
- If the distance is less than a predefined threshold (indicating an obstacle is close), the microcontroller decides on the appropriate action to avoid the obstacle.
- Based on the distance data, the microcontroller sends signals to the motor driver.
- The motor driver then controls the DC motors to change the bot's direction or stop it.
- If no obstacle is detected, the bot continues moving forward.

THANK YOU!